

Lecture 4 (153-248)

Multiple Linear Regression

→ y : Response

→ $x_1, \dots, x_m$: Covariates

Data: $(y_i, x_{i1}, x_{i2}, \dots, x_{im})$
 $i = 1, \dots, n$

Response vector: $y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \rightarrow n \times 1$

Covariate matrix:

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & \dots & x_{1m} \\ \vdots & x_{21} & x_{22} & & x_{2m} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & & x_{nm} \end{bmatrix}$$

$\downarrow$
 $n \times (m+1)$

Time Series: $y_t, t = 1, \dots, n$

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

① Use functions of time to construct x .

$$y_t = \boxed{\beta_0 + \beta_1 t + \beta_2 t^2} + \text{error}$$

$$X = \begin{bmatrix} 1 & 1 & 1 \\ \vdots & \vdots & \vdots \\ 1 & n & n^2 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & \cos \frac{2\pi t}{T_2} & \sin & \cos(\) \\ \vdots & \vdots & \vdots & \vdots \\ 1 & \vdots & \vdots & \vdots \end{bmatrix}$$

② Use lagged values of y as covariates.

$$X = \begin{bmatrix} y_{t-1} & y_{t-2} & \dots & y_{t-m} \\ \vdots & \vdots & \vdots & \vdots \\ y_{n-1} & y_{n-2} & \dots & y_{n-m} \end{bmatrix}$$

$$y_1, \dots, y_n \quad m=2$$

$$y = \begin{pmatrix} y_3 \\ \vdots \\ y_n \end{pmatrix} \quad X = \begin{bmatrix} 1 & y_2 & y_1 \\ \vdots & \vdots & \vdots \\ 1 & y_{n-1} & y_{n-2} \end{bmatrix}$$

$$\boxed{y, X}$$

$$sm.OLS(y, X)$$

Frequentist Inference

$$y_i = \underbrace{\beta_0}_{\text{intercept}} + \underbrace{\beta_1}_{\text{slope}} x_{i1} + \dots + \underbrace{\beta_m}_{\text{slope}} x_{im} + \underbrace{\varepsilon_i}_{\text{error}}$$

unknown parameters:
 $\beta_0, \beta_1, \beta_2, \dots, \beta_m, \sigma$

$$\rightarrow \varepsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$$

ε_i mean 0
independence
finite variance

- ① Construct estimators for the parameters. MLE, least squares
- ② Figure out the distribution of the estimate.
(exact or approximate)
Use the distribution to construct confidence intervals.

Estimate: Least Squares

$$\beta = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_m \end{pmatrix}$$

$\beta_0, \dots, \beta_m$

$$S(\beta) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_m x_{im})^2$$

minimize $S(\beta)$ to get $\hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \vdots \\ \hat{\beta}_m \end{pmatrix}$

Solve $\hat{\beta}$ in closed form:

$$\begin{aligned} S(\beta) &= \|y - X\beta\|^2 \\ &= (y - X\beta)^T (y - X\beta) \\ &= y^T y - \underbrace{y^T X \beta}_{\text{cross term}} - \underbrace{\beta^T X^T y}_{\text{cross term}} + \beta^T X^T X \beta \end{aligned}$$

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

$$X = \begin{pmatrix} 1 & x_{11} & \dots & x_{1m} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{nm} \end{pmatrix}$$

$$\nabla S(\beta) = \begin{pmatrix} \frac{\partial S}{\partial \beta_0} \\ \vdots \\ \frac{\partial S}{\partial \beta_m} \end{pmatrix} \quad f(x_1, x_2) = a_1 x_1 + \dots + a_n x_n$$

$$\nabla f = a$$

$$\nabla S(\beta) = -x^T y - x^T y + 2x^T x \beta \quad (\text{check})$$

$$\nabla (\beta^T x^T x \beta)$$

$$\nabla (\beta^T A \beta) = 2A\beta \text{ if } A \text{ is symmetric.}$$

$$\nabla S(\beta) = -2x^T y + 2x^T x \beta = 0$$

$$x^T x \beta = x^T y$$

$$\Rightarrow \hat{\beta} = (x^T x)^{-1} x^T y$$

Least Squares Estimator.

Maximum likelihood.

$$\varepsilon_i \text{ iid } N(0, \sigma^2)$$

$$\rightarrow y_i \text{ independent } \sim N(\beta_0 + \beta_1 \varepsilon_1 + \dots + \beta_m \varepsilon_m, \sigma^2)$$

Data: $(y_i, x_{i1}, \dots, x_{im})$

Assume x_{ij} 's are fixed.

$$\prod_{i=1}^n \frac{1}{\sqrt{2\pi} \sigma} \exp \left[-\frac{(y_i - \beta_0 - \dots - \beta_m x_{im})^2}{2\sigma^2} \right]$$

$$= (2\pi)^{-n/2} \sigma^{-n} \exp \left[-\frac{S(\beta)}{2\sigma^2} \right] \rightarrow \text{Likelihood}$$

log-likelihood

$$= -\frac{n}{2} \log 2\pi - n \log \sigma - \frac{S(\beta)}{2\sigma^2}$$

$$\nabla S(\beta) = 0 \Rightarrow \hat{\beta}_{MLE} = \hat{\beta}_{\text{Least Squares}}$$

$$-\frac{n}{\sigma} + \frac{S(\beta)}{\sigma^3} = 0 \Rightarrow \sigma = \sqrt{\frac{S(\beta)}{n}} \rightarrow$$

$$\hat{\sigma}_{MLE} = \sqrt{\frac{S(\hat{\beta})}{n}}$$

Remember: $\hat{\beta}_{MLE} = \hat{\beta}_{\text{least squares}} = \hat{\beta} = (X^T X)^{-1} X^T y$

$$\hat{\sigma}_{MLE} = \sqrt{\frac{S(\hat{\beta})}{n}}$$

Step 2: Calculate the distribution.

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

$$y_i \stackrel{\text{ind}}{\sim} N(\beta_0 + \beta_1 x_{i1} + \dots + \beta_m x_{im}, \sigma^2)$$

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \sim N$$

Recall: $x \sim N_p(\mu, \Sigma) : \frac{1}{(\sqrt{2\pi})^p \sqrt{|\det \Sigma|}} \exp\left(-\frac{(x-\mu)^T \Sigma^{-1} (x-\mu)}{2}\right)$

①

$$\textcircled{b} \quad X \sim N(\mu, \Sigma) \Rightarrow AX \sim N(A\mu, A\Sigma A^T)$$

$$\hat{\beta} = \underbrace{(X^T X)^{-1} X^T y}, \quad \boxed{y \sim N(X\beta, \sigma^2 I)}$$

$$y_i \stackrel{\text{ind}}{\sim} N(\underbrace{\beta_0 + \beta_1 x_{i1} + \dots + \beta_m x_{im}}_{X\beta}, \sigma^2)$$

$$\underbrace{\begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}}_y \sim N \left(\underbrace{\begin{pmatrix} \beta_0 + \beta_1 x_{11} + \dots + \beta_m x_{1m} \\ \beta_0 + \beta_1 x_{21} + \dots + \beta_m x_{2m} \\ \vdots \\ \beta_0 + \beta_1 x_{n1} + \dots + \beta_m x_{nm} \end{pmatrix}}_{X\beta}, \underbrace{\begin{pmatrix} \sigma^2 & & 0 \\ & \ddots & \\ 0 & & \sigma^2 \end{pmatrix}}_{\sigma^2 I_n} \right)$$

$$\left. \begin{array}{l} y \sim N(X\beta, \sigma^2 I) \\ \hat{\beta} = \underbrace{(X^T X)^{-1} X^T y}_A \end{array} \right\} \Rightarrow \boxed{\hat{\beta} \sim N(\beta, \sigma^2 (X^T X)^{-1})}$$

DETAILS
SKIPPED.

$$\boxed{\hat{\beta} \sim N(\beta, \sigma^2 (X^T X)^{-1})}$$

$$\hat{\sigma}_{MLE} = \sqrt{\frac{S(\hat{\beta})}{n}}$$

$$\Rightarrow \hat{\sigma}_{MLE}^2 = \frac{S(\hat{\beta})}{n} \sim \frac{\sigma^2}{n} \chi_{n-m-1}^2$$

$$S(\hat{\beta}) = \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - \dots - \hat{\beta}_m x_{im})^2$$

$$\hat{\sigma}_{MLE}^2 \sim \frac{\sigma^2}{n} \chi_{n-m-1}^2$$

$$\mathbb{E} \hat{\sigma}_{MLE}^2 = \frac{\sigma^2}{n} \times (n-m-1)$$

$\hat{\sigma}_{MLE}^2$ is NOT unbiased for σ^2 .

$$\hat{\sigma}_{unbiased}^2 = \frac{S(\hat{\beta})}{n-m-1} \leftrightarrow \hat{\sigma}_{MLE}^2 = \frac{S(\hat{\beta})}{n}$$

Recap: $\hat{\beta} = (X^T X)^{-1} X^T y \sim N(\beta, \sigma^2 (X^T X)^{-1})$

$$\hat{\sigma}_{MLE}^2 \sim \frac{\sigma^2}{n} \chi_{n-m-1}^2$$

$$\hat{\sigma}_{unbiased}^2 = \frac{n}{n-m-1} \hat{\sigma}_{MLE}^2$$

$$\hat{\beta}_j \sim N(\beta_j, \sigma^2 (X^T X)^{-1} (j+1, j+1))$$

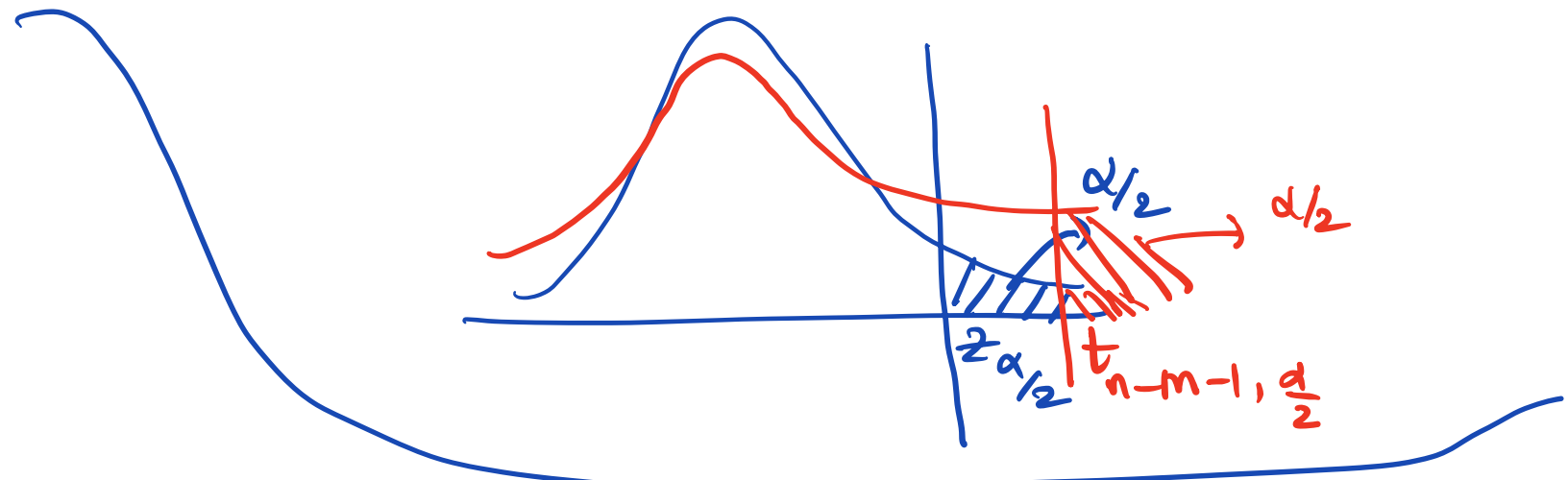
$$\left[\hat{\beta}_j - z_{\frac{\alpha}{2}} \sigma \sqrt{(X^T X)^{-1} (j+1, j+1)}, \hat{\beta}_j + z_{\frac{\alpha}{2}} \sigma \sqrt{(X^T X)^{-1} (j+1, j+1)} \right]$$

$$\left[\hat{\beta}_j - z_{\frac{\alpha}{2}} \hat{\sigma}_{unbiased} \sqrt{(X^T X)^{-1} (j+1, j+1)}, \hat{\beta}_j + z_{\frac{\alpha}{2}} \hat{\sigma}_{unbiased} \sqrt{(X^T X)^{-1} (j+1, j+1)} \right]$$

t-distribution quantile $t_{n-m-1, \frac{\alpha}{2}}$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

$$y \sim N(X\beta, \sigma^2 I)$$



Bayesian Approach

Model:

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_m x_{im} + \varepsilon_i$$

$i=1, \dots, n$

$\varepsilon_i \sim N(0, \sigma^2)$

$[m+1+n]$ variables

$\beta_0, \dots, \beta_m \sim \text{Unif}(-C, C)$

$\log \sigma \sim \text{unif}(-C, C)$

$$(\beta_0, \dots, \beta_m) \mid \text{data } y_1, \dots, y_n \quad \sigma \mid \text{data } = (y_1, \dots, y_n)$$

$$f_{\beta | \text{data}}(\beta)$$

$$f_{\beta, \sigma | y_1, \dots, y_n}(\beta, \sigma)$$

$$\propto \underbrace{f(y_1, \dots, y_n | \beta, \sigma)}_{\text{likelihood}} \underbrace{f_{\beta, \sigma}(\beta, \sigma)}_{\text{prior}}$$

$$\Rightarrow \sigma^{-n} \exp\left(-\frac{S(\beta)}{2\sigma^2}\right) \frac{I(-C < \beta_j < C)}{(2C)^{n+1}} \frac{I(-C < \log \sigma < C)}{2C\sigma}$$

$$\propto \sigma^{-n-1} \exp\left(-\frac{S(\beta)}{2\sigma^2}\right) I(\sigma > 0)$$

$$f_{\beta | \text{data}}(\beta)$$

$$\propto \int_0^\infty \sigma^{-n-1} \exp\left(-\frac{S(\beta)}{2\sigma^2}\right) d\sigma$$

$$\propto \left(\frac{1}{S(\beta)} \right)^{n/2}$$

$$\left(\frac{S(\hat{\beta})}{S(\beta)} \right)^{n/2} \rightarrow t\text{-distribution.}$$